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RISK CLASSIFICATION AND RATEMAKING IN INSURANCE

A Thesis

Submitted

in Partial Fulfillment

of the Requirements for the Designation

University Honors

Erin Conrad

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Abstract

Actuaries in insurance companies strive to put a price tag on future risk. They use a variety of analyses and models to accurately price insurance products. Models discussed in this paper include the pure premium method, the loss ratio method, the minimum bias procedure, and generalized linear models. Each of these pricing mechanisms are used to determine accurate and actuarially sound rates for insurance products. Competition in the industry forces companies to find more refined ways to predict the future costs of a risk. Classification systems allow companies to include a variety of rating variables so that each risk is priced based on its individual characteristics. This paper explains the basics of ratemaking as well as describes each of the models mentioned. It will also look at the advantages and disadvantages of each model and determine when one model is preferable to others.

1. INTRODUCTION

There are two fundamental goals of insurance companies: ratemaking and loss reserving. Ratemaking includes creating models and putting an accurate price tag on future risks. Loss reserving is setting aside enough money to cover the risks to which a company is exposed. The costs of these risks are unknown and must be predicted using past experience. Using past data actuaries create models to evaluate the correct prices to charge insureds. The predicted costs are then used to create rates for individuals and policies. The process of creating rates and other risk transfer procedures is called ratemaking. The main goal of ratemaking is to determine appropriate and actuarially sound rates for individual insureds and policies. Each company decides how they are going to create rates and price individual risks using pricing plans.

Competition is strong in the insurance industry. Companies are striving to find the best way to predict future costs of individual risks. Insurance products such as homeowners' insurance and automobile insurance are very common in households today. Many adults purchase these products to protect against potential losses. Therefore, there is a lot of data that can be analyzed and used in the ratemaking process. Credibility is used to determine how well the model fits the data based on the reliability of the data for determining rates. Companies use their data to determine the best price for a risk based on the individual characteristics of that risk. Large companies have more access to complete data sets making credibility less of an issue. With these large data sets, the risks can be broken down into smaller groups and rates can be based on the group's attributes. However, the groups must have enough data to be credible. If groups of data are too small, they will not have enough credibility to produce actuarially sound

rates. The data must be split into groups where there is sufficient information for the basis of the rates. There are an unlimited number of ways for a company to do this.

Each company may choose to split their data in different ways on the basis of rating variables. Rating variables are the factors upon which an individual rate or price is based. For example, a company may group their risks for automobile insurance based on age and gender. They may charge young males more than they charge older females. A second company may also choose to include the type of car a person drives in their rates. This allows companies to differentiate their products and create more accurate rates. Other products may not have characteristics that can be as easily classified. In addition, they may have a smaller set of data to be used for predicting future costs. For these rates, other ratemaking techniques may need to be used since it is not possible to split the data into groups and maintain credibility. Some products price individual rates separately while others may be thrown into large groups and a price created for the aggregate. Using these various ratemaking techniques, companies try to gain a competitive advantage by pricing their risks with the best model and with as much accuracy as possible. If they are able to charge safer risks lower prices and more hazardous risks higher prices, then the company will be able to gain customers and market share. When the company's rates are too high, they will lose business. When the company's rates are too low, they will gain market share, but there is a potential for large losses and higher exposure to hazardous risks. Actuaries perform in-depth analyses to predict the future costs and risks to the company of each individual insured so that they can accurately price products to make their company profitable.

Ratemaking is used mainly in property and casualty insurance. In homeowners' insurance, rates are created based on location, size, building material, and many other rating variables. Actuaries strive to create prices for these insurance products that cover the risks and

compete within the industry. Finding the best way to price the product enables the company to gain customers and market share. Classification plans also attempt to reduce rates for the better risks and increase rates for more hazardous risks. Differentiating prices between risks creates a more effective and efficient pricing structure. The effectiveness of the plan is how well the plan separates the better risks from the overall average (Bailey & Simon, 1960). This goal allows companies to price insureds according to their relative risk in the industry, and thus, try to create a competitive advantage in the market for their product.

Various models can be used in ratemaking. The pure premium method calculates the predicted cost for the insured and determines the price that should be charged including profit and other costs. The loss ratio method uses a target loss ratio to calculate the price to charge an insured. Risk classification is used to put a particular insured into a class and then determine the rate for the group of insureds. With the expanding use of predictive modeling, procedures such as the minimum bias model and generalized linear models are becoming more common in pricing insurance products. These procedures use multiple classification dimensions to price products such as automobile insurance and homeowners insurance. The rates produced by these processes can be based on various factors creating an opportunity to differentiate prices between customers and risks based on their individual attributes.

This paper will explain the basic ratemaking concepts and various ratemaking models. There are many different ways to price insurance products. However, not all models can be used in every situation. Using the pure premium approach or loss ratio approach does not allow a company to distinguish between the characteristics of individual risks. Since one group may be at a higher risk than another group, they should be charged more for their policy. Using the minimum bias model or generalized linear models, actuaries are able to classify risks based on

their individual characteristics. The individual characteristics allow actuaries to create more refined classification plans. In choosing which model to use, an important factor is credibility. For the model to be a good estimate of future risks, an actuary must have plenty of data that is accurate and reliable. This paper will explore the advantages and disadvantages of various pricing models and explain when each model is appropriate for use in pricing insurance. The minimum bias procedure and generalized linear models allow actuaries to create pricing models that can be used to price any number of risks. The pure premium method and the loss ratio method can be used only when rates need to be changed for one group of risks. Determining the appropriate method is one of the most important aspects of the ratemaking process.

2. RATEMAKING CONCEPTS

Ratemaking is the process of creating rates or other risk transfer mechanisms within the insurance industry. The main goal is to create appropriate rates for individual insureds by predicting the costs associated with the individual. There are many terms associated with losses and ratemaking. The actual losses for a policy at a specific point in time are incurred losses (Casualty Actuarial Society, 1988). These losses can increase or decrease over time due to changes in the loss amounts predicted. The initial prediction could be significantly higher or lower than what the claim actually settles for. Paid losses are the losses that have been paid out during a specific time period. These losses tend to increase over time. It may take months or years before claims can be paid and settled. For asbestos claims, it takes lots of years for the claims to be paid. However, for automobile insurance claims, it may take only a few months. Before these losses can be used in ratemaking they must be developed to their ultimate value.

Historical data is used to determine how losses for a specific type of insurance develop over time (McClenahan, 2001). Development factors are applied to the paid losses to calculate the predicted ultimate losses. Then the losses can be used in analyses for pricing products.

Ratemaking also must consider the expenses associated with pricing the risks and paying out the claims. Allocated loss adjustment expenses are settlement costs that can be directly associated with a specific claim. These may include expenses such as legal costs and medical containment costs. Unallocated loss adjustment expenses are those that cannot be directly associated with a specific claim. For example, adjusters' salaries, claim department expenses, and staff attorney's fees are not directly attributed to specific claims. Commission and brokerage expenses include the money paid to agents and brokers. Other costs include taxes, licenses, and fees as well as general administrative expenses. These costs all must be included in ratemaking to accurately price an insurance product (Casualty Actuarial Society, 1988). With the addition of profit, these expenses and the pure premium make up the basis for the rate charge to an individual insured. It is important to include all costs in calculating the final premium to charge an individual.

Ratemaking includes three principles. First, the rate must be an estimate of the expected value of future costs since policies are priced before the losses occur. For example, when an insured pays the premium for their automobile insurance policy, the actual costs associated with that insured are unknown. The claims could end up being significantly more or less than what the insured actually paid. Secondly, the rate must cover all costs that are associated with a transfer of risk. The costs for each risk include the actual claim expenses, profit, and all of the expenses mentioned above. Lastly, the rate must provide for all costs associated with an individual risk transfer to produce an actuarially sound rate. An actuary's goal is to create a rate

that is reasonable and not excessive, inadequate, or unfairly discriminatory. One factor that is not commonly used in pricing products is race since it could be considered unfairly discriminating to certain groups of people. If these three principles are considered, then the rates that are created will constitute an actuarially sound rate that is the best estimate of the expected value of the costs that are associated with an individual risk transfer (McClenahan, 2001).

In ratemaking, a pure premium must be calculated before the price to charge the insured is determined. This pure premium is equal to the total loss divided by the number of exposure units. It does not include any costs associated with the policy or claim. The pure premium is usually calculated using historical claims or losses. In addition, a loss ratio can be calculated for an individual policy, group of policies, department, or company as a whole. The loss ratio is the total loss divided by the total premium. This loss ratio tells a company how well the premiums they are collecting are covering the total losses. The amount an individual actually pays to be covered by an insurance policy is the premium. The premium charge to an individual is usually based on the number of exposures. The characteristics of the individual risk are taken into account through the use of rating variables, or rating factors. These rating variables make up the structure of the rating plan and are included in the manual rating of specific risks. Rating plans allow the rates that are charged to reflect the identified losses of the insureds (McClenahan, 2001). When an insured demands a payment under the insurance contract, they are placing a claim. These claims can be made by the insured themselves or by a third party. When there is a car accident, the person who caused the accident makes a claim for their vehicle and personal medical expenses. The insurance company for the other person also makes a claim for the physical damage to the other vehicle and medical expenses for that person. All of these terms are used in the ratemaking process and must be considered to produce accurate rates.

3. EXPOSURE BASES

Insurance premiums are based on the basic rating unit called an exposure (McClenahan, 2001). The exposure base is used to calculate the manual premium, or the amount the insured should be charged. It is important that exposures are as closely proportional to costs as possible (Finger, 2001). For example, in automobile insurance, one automobile being insured for a period of twelve months is considered a car year. The number of exposures written during a specific period of time are called written exposures. They are most commonly used in analyses, but earned exposures can be used as an alternative. Earned exposures are those exposure units actually exposed to losses during the specific time period. For example, if only six months have passed then the earned exposures are one half of a car year. In addition, in-force exposures are the exposure units at a given point in time.

Exposure units are chosen based upon reasonableness, ease of determination, responsiveness to change, and historical practice. It is very important that the exposure unit relates to loss potential. Ease of determination is important because if the exposure unit cannot easily be determined then it is of no use. For example, the number of cars a family has is both a reasonable exposure unit and easy to determine because the cars can be counted. The number of miles driven in a year is less easily determined since it may vary from year to year and person to person. Exposure units can change over time which means that they must be responsive to change. In workers' compensation insurance, the exposure base is typically payroll. Payroll is responsive to change since it varies as employees enter and exit the organization (McClenahan, 2001). Historical bases may also be used for the purposes of estimating future risk

(McClenahan, 2001). Historical bases include claim history and past payroll. These could be used to predict the future exposures of an organization. It may be important to look at trends or other factors that cause exposures to increase or decrease over time. Another example is to use the number of cars as the exposure base and the claim history for those policies. These historical costs can be used to predict future claims by the insured and then assist in the pricing of policies. If historical exposure bases are to be used, the actuary must believe that the historical bases are a good predictor of future bases.

4. RATING VARIABLES

Once the exposure bases are chosen, an actuary must consider the rating variables that are to be used in the ratemaking process. Rating variables are the factors that are used in creating the model and determining the correct price to charge and individual. They can also be called rating factors. Choosing reliable and appropriate rating variables is important for creating a trustworthy and accurate model. If actuaries do not choose the correct rating variables, then they will not be able to create an adequate model. This could cause the company large losses or in extreme cases, it could force them out of business. There are four criteria for selecting the rating variables: actuarial, operational, social, and legal (Finger, 2001).

The statistical measures including accuracy, homogeneity, credibility, and reliability are the actuarial criteria. In selecting rating variables, the most important of these four is accuracy. The rating variables must be accurate for companies to profit and continue to operate. Accurate rating variables are also important for creating fair rates. When insurers are able to price products more accurately, they are usually more successful in competitive environments. In

general, more rating variables and a more detailed the classification system create more accurate rates. However, too much detail can make selling the product more tedious. Random fluctuations in claim experience make estimating costs for small groups or individuals very difficult (Finger, 2001). The concept of credibility is that groups must be large enough to measure costs with accuracy. Therefore, cutting the data down to very small groups can cause problems in the pricing plan. When members in a group have the same expected costs and receive the same rate or premium, the group is said to be homogenous. However, it is difficult to know what the expected costs are for each individual, and therefore, can be difficult to create homogenous groups. Insurers try to group insureds by their characteristics to create homogenous rates for groups of people. Reliability and predictive stability are important since costs may change over time. Once again, these differences could be due to random fluctuations (Finger, 2001). For example, if the relative risk of males to the base rate in automobile insurance changes over time, then the rates may no longer be reliable. Changing risks cause rates to become unreliable as time progresses. Therefore, there must be continuous analyses performed to ensure that current rates are accurate and fair.

Operational considerations are also important in determining the appropriate rating variables. Rating variables should have an objective definition, and there should be little ambiguity. Classes should be mutually exclusive and exhaustive (Finger, 2001). This makes the rating plan less confusing and keeps insureds from manipulating the group they are in, and thus, the premium they pay. For example, if an individual can change their premium by saying that they will drive their car in rural areas rather than urban areas, then they are able to manipulate their price. The categories of rural and urban may not be mutually exclusive since many people drive in both areas. Thus, in order for them to be used in a pricing plan, specific definitions must

be created so that the categories are mutually exclusive and manipulations cannot be made. Urban and rural are factors used in automobile insurance today. Insurers should also be able to verify their data. This prevents insureds from being dishonest to achieve a lower price. Rating variables that are free from manipulation include gender, age, and many others. The costs of obtaining and verifying information may also need to be taken into consideration. Administrative expense may be too high when the cost of obtaining information exceeds the value of the additional accuracy. There may come a point in an analysis when adding another variable is more expensive than the benefit and pricing accuracy actually gained from it. Therefore, the rating variable is not added because it is not easily verified. Lastly, practical considerations include that the variables need to be related to the costs (Finger, 2001). For example, age of the individual is an important variable in life and automobile insurance while it probably has no real value in creating rates for homeowners' insurance. The age of the home may be a better predictor of the risk for homeowners' insurance. Choosing rating variables that are unambiguous and free of manipulation are important operational considerations.

Social considerations, including causality, controllability, and affordability, must also be taken into account when choosing rating variables. Causality can result when the mean costs for various groups are significantly different. The rating variable may be correlated with the costs. Variables that are under control of the insured are called controllable variables. This allows insureds to act in a certain way to reduce their premiums. This could be a potential problem since risky insureds can manipulate their premium to be less than the costs of their actual risk. However, it could also encourage accident prevention. The number of fire detectors in a house is easily controlled by an individual, but if they have more fire detectors then there may be less chance of extensive fire damage costs. In addition, administrative costs may be increased if the

insured has control over their premium costs. The insurer may require validation to determine the actual premium resulting in higher expenses to the company. If classification systems are more refined, affordability may also become a concern (Finger, 2001). High rates make it difficult for some people to get insurance. For example, many states require automobile liability insurance, but if rates are too high, then some people will be unable to afford it. Lastly, since some people are hesitant to release personal information, privacy can be a concern. The reluctance to disclose personal information affects the accuracy of the classification, verifiability, and administrative costs. Rates must be socially acceptable for the general public to understand and accept them.

Legal considerations vary widely between cities, states, and regions. However, they have significant effects of risk classification. It is important for insurance companies to be aware of regulations before writing business in particular areas. Many states set limits on the amount that a company can charge for a certain type of insurance. Constitutions, statutes, and regulations all must be considered before pricing insurance products. These considerations may cause companies to charge less for products than what the actuaries calculate to be appropriate rates. In fact, companies generally charge less than actuaries calculate to be sufficient due to the need to write business, be competitive, and appeal to customers. In this case, if insurers write business in a region and charge lower rates than they think are adequate, they may be setting the company up for failure and large losses. Therefore, they must decide whether they are going to write business in that particular location. They may decide that they are not able to charge enough for their products, and therefore, it is too risky of a business. If the company does not think they can appropriately cover their costs, then they will be unable to earn a profit. Writing business in an area and charging less than actuarially calculated could ultimately cause a company to go out of

business. In addition, sometimes insurers are not able to use particular variables to price products. Therefore, insurers must find replacement variables or potentially leave the market because removal of the rating variable reduces the accuracy of the classification system (Finger, 2001). Some people think it is unfair to base automobile insurance premiums on gender. In general, males are more risky and charged more than a female with the same characteristics. If insurers are unable to use gender as a rating variable, then they will charge females too much and males not enough. Therefore, the insurers would try to find a replacement variable for gender, or they may even choose to leave the market. Legal considerations vary between state and region and are sometimes difficult to account for since they usually limit the amount a company can charge for a product.

After considering the criteria for determining rating variables, insurers try to find the best way to rate their products and make price differentials. Therefore, classification systems have varied over time. Competition between companies forces them to strive for the best rating plan possible. As coverage becomes more expensive and markets become larger, classification systems are becoming more refined. Information technology has allowed classification systems to increase in complexity. Computers and software programs are able to process information quicker and in a more cost effective way (Finger, 2001). Predictive model is becoming more widely used in property and casualty insurance with the increasing power of technology. Using technology, companies are able to price insurance products with more accuracy and according to more complex models to boost their profits. This ultimately leads to a much greater need to find and choose the best rating variables possible to be competitive.

5. RATEMAKING OBJECTIVES

Ratemaking includes a variety of essential objectives. The first is to produce a rate that covers expected losses and expenses. A company wants their income to equal the money that they are paying out in expected losses and expenses plus a profit. Actuaries must also produce rates that include a provision for unexpected losses. For example, a large hurricane occurs very seldom, but actuaries must take this into account when they are pricing products. A third objective is to encourage loss control. This gives the policyholders an incentive for reducing their risk. One example of a loss control is a good driver discount in automobile insurance. Lastly, the rates must satisfy rate regulators. Rates must be adequate to cover expenses but not excessive. They also cannot be unfairly discriminatory (Brown & Gottlieb, 2001). Regulators have the final say in the rates that can be charged.

There are also some desirable but not essential objectives. First, rates should be reasonably stable. If rates are not stable, the general public buying the insurance may start to question them. Small increases are expected from time to time due to inflation. However, if rates jump up and down from year to year customers may become confused and angry. Secondly, rates should be responsive to change. This lies in choosing a base exposure that is responsive to change. In workers' compensation, payroll is an exposure base that is responsive to change. Lastly, rates should be simple and easy to understand. This allows the policyholder to understand how loss control relates to lower rates. It also helps the policyholder to see that the rates are legitimate and fair. In addition, agents and brokers may have a hard time understanding complex classification systems leading them to encourage customers to purchase insurance from other companies (Brown et al., 2001). For example, in homeowners insurance, if a company

uses rating variables that are hard to measure, such as how much a fireplace is used, there is ambiguity for agents and brokers. They may choose to use less ambiguous classification systems since the pricing plan is easier to use. The company with the complex plan loses out on potential customers and income because of the price associated with convenience.

If all of these objectives are met when rates are created, the rating system is more likely to be accepted by policyholders. Agents and brokers can easily implement the system. The risk that the company has accepted is covered with the premium paid by the insureds. If companies do not follow these objectives, they may experience large losses and not have enough to cover them. The company could also lose business, and their profits could be reduced. If this happens over and over, the company may ultimately be forced out of business.

6. RISK CLASSIFICATION

When there is not enough information to estimate the price for a given insured, risk classification must be used. Individuals with similar expected risks are grouped together. The overall price for the group is calculated and then divided by the number in the group to determine the premium to charge each individual. Risk classification is defined as the formulation of different premiums for the same coverage based on group characteristics, or rating variables (Finger, 2001). The variation of the actual costs from the expected costs is the risk. Grouping the individuals allows for more accurately priced products rather than charging everyone the same. This makes the estimates more credible than if the costs were calculated for an individual insured.

Sometimes an individual's claim experience may have enough credibility to justify modifying the rate to charge the individual. For very simple classification plans, the credibility of the individual claim experience is greater. However, for more complex classification plans, the credibility of the individual claim experience is less. In general, a more accurate classification plan is preferred to individually rating the risks (Finger, 2001). Underwriters are given the ability to credit and debit base rates based on the classification plan to determine the overall rate to charge the insured. The practice of crediting and debiting base rates is used for insurance products such as property insurance for an entire city or a university.

A variety of strategies are used by insurers to price products. Marketing and underwriting techniques are adjusted to account for various factors. The price can be adjusted according to the individual costs potential. A company may also choose to accept an insured only if the existing price structure is adequate to cover the costs. In addition, there may be several different rate structures for an insurance group. Insurers may be required to have different companies due to rate laws (Finger, 2001). One company may charge 20% more, one 10% more, and one the base price while a fourth company charges 10% less than the base rate. The underwriter uses the information available to them to determine which company an individual insured falls into. Some insurance companies create checklists for this purpose. A scale is used to determine the company the risk is place in depending on how the risk meets the criteria. This company determines how much the underwriter debits or credits the premium.

Determining how well classification systems are working is very important. The measure of the accuracy of a classification system is the efficiency. The reason for having classification systems is to assess the variability in costs from one insured to another. Understanding this variability is the key to measuring the efficiency of the system (Finger, 2001). In a perfect

classification system, the system would have the same variability of the insured population since all insureds would be properly classified, but this is not possible because costs are unknown. Class relativities and the percentage of exposures in each class are needed to determine variability. The variability is unknowable since costs are dependent on future events. However, concepts of risk theory can be applied to calculate estimates. The validity of estimates depends on a variety of factors including the chosen rating variables and the accuracy of the data. The sample data must be representative of the costs that are expected, and the process variance must be sufficient for calculating the unknown variability (Finger, 2001).

Risk classification uses group characteristics to formulate the premium. Ultimately the task is to price an individual insured's risk. However, their claim data may be insufficient. Because of this, group costs can be used, and the assumption is made that an individual belongs to a particular group. The groups must be relatively homogenous, sufficiently large, and reasonably stable over time. In competitive markets, classification systems are more refined and premiums are more accurate (Finger, 2001).

7. CREDIBILITY

Credibility is assessing the relative meaningfulness of a group's costs versus the meaningfulness of the credibility complement's costs. There are two problems related to credibility. The first lies in how to obtain more data or more reliable data, and the other is determining the most appropriate credibility complement. The first problem can be fixed by using more years of data or data from several states (countrywide). However, countrywide data may not a reliable estimate for an individual state (Finger, 2001). For example, coastal states

have a much higher risk of hurricanes than interior states. Therefore, they may need to be charged more for homeowners' insurance than what the countrywide average would indicate. In this case, it is better to focus on the smaller area and then make an adjustment for credibility. Relying on historical costs could also be a problem for credibility. If the historical costs cannot be used to predict the future costs, then the data is not fully credible. The credibility of each method depends on the amount of data that the company has as well as the accuracy of the data they are using. For rates to be accurate, companies must have enough data to perform analyses that are believed to give credible estimates of future costs. This is usually based on analyses of past data and costs.

8. PURE PREMIUM METHOD

One of the two very simple methods for ratemaking is the pure premium method. The average loss per unit of exposure is called the pure premium. The pure premium is calculated by dividing the total loss by the exposure units. It can also be calculated as the product of the frequency per unit of exposure and severity (McClenahan, 2001). The pure premium method is most common when earned exposures are used as the exposure base (Finger, 2001).

The pure premium method calculates a rate that includes the expected losses, expenses, and profit. The formula for calculation of the pure premium is:

$$R = \frac{P + F}{1 - V - Q}$$

where

R = rate per unit of exposure
P = pure premium
F = fixed expense per exposure
V = variable expense per exposure
Q = profit and contingencies factor

The pure premium method requires a significant amount of information. Therefore, it is usually rather accurate for individual groups. Another advantage is that frequency and severity relativities can be produced as well as pure premium relativities (Finger, 2001). The frequency and severity relativities could then be used to individualize prices. Actuaries may use the pure premium method to calculate premiums for simple types of insurance. It does not take into account various rating variables and is less flexible. Therefore, the pure premium method would not be effective in calculating rates for automobile insurance or homeowners' insurance. The pure premium method is most appropriate for simple classification systems such as those used for workers' compensation insurance. The pure premium method may also be useful when changing rates for one specific group of insureds. Rather than recalculating rates for all automobile insurance policies, a company may decide to revise rates only for a specific group allowing them to use the pure premium method. Therefore, the pure premium method is most useful for specialized types of insurance when it is not realistic to classify risks by characteristics.

9. GENERAL PURE PREMIUM MODEL

The general pure premium method is a bit more complicated. One again the pure premium is calculated by dividing total losses by the number of exposures. Then an actuary must determine what rate to charge the policyholder. This rate must include all costs that are associated with the individual transfer of risk. The premium charged to the policyholder is equal to the number of exposures times the rate calculated. Expenses such as commission and brokerage, taxes, licenses, and fees, other acquisitions, general expenses, and pretax profit must be included in the rate. Each of these five categories are separated into variable and fixed costs in the general pure premium model. The model consists of the sum of the component parts: the pure premium, fixed expenses, variable expenses, and a provision for profit and contingencies. The more general model of the pure premium method as proposed by Schofield (1998) is:

$$R = P + f(E) + v(R,E) + Q \cdot R$$

where

R = rate per unit of exposure
P = pure premium (expected loss cost) per unit exposure
E = number of written exposures on the policy
f(E) = fixed expenses as a function of total written exposures
v(R,E) = variable expenses as a function of the rate and written exposures
Q = profit and contingencies factor (as a proportion of the rate)

First, the actuary must be decided whether fixed expenses are allocated to a policy or an exposure. If the rate is on an exposure basis, then the fixed expenses calculated on a policy bases must be converted to an exposure basis. When fixed expenses for a policy are constant, they vary for exposures. However, if fixed expenses for an exposure are constant, then they vary for each policy. This is due to the fact that the premium for a policy is the number of exposures times the rate. The per exposure basis is used most often when the range of exposures written on

a policy is small. The per policy basis is used when a wider range of exposures can exist on an individual policy (Schofield, 1998). For example, workers' compensation is an example of a type of insurance that has a wide range of exposures for each policy meaning that a per policy basis is usually used.

The first method of determining the rate to charge is the expense fee method. This includes the amount that is paid to cover the costs of fixed expenses that are associated with writing and servicing a policy. The general equation for this method is:

$$R = P + F + V \cdot R + Q \cdot R$$

where

F = fixed expenses per exposure
H = fixed expense ratio (fixed expenses as a proportion of the rate)
V = variable expense ratio (variable expenses as a proportion of the rate)

The expense fee method reduces to the pure premium method above that was introduced by McClenahan (2001). R is independent of the number of exposures written for the policy. Each exposure pays the same amount of fixed expense.

A second method is the workers' compensation method where the fixed and variable expenses are allocated based on a per policy basis. Each policy pays a fixed amount to cover the fixed expenses associated with the policy. The function *f(E)* takes the following form:

$$f(E) = \frac{Policy\ Expense\ Constant}{E}$$

The expenses that are not covered by the expense constant are either variable or a non-constant function of the exposures or premium. As the premium increases, commission and general expense ratios decrease. Therefore, this method gives policyholders with large policy premiums a discount. Not all expenses are directly proportional to the premium or rate (Schofield, 1998).

In general, the pure premium method is used to calculate a rate based on the loss per exposure unit and expenses associated with it.

10. LOSS RATIO METHOD

The second very simple ratemaking approach is the loss ratio method. The loss ratio method is typically used to determine rate changes rather than the rates themselves (McClenahan, 2001). When earned premium is used, the loss ratio method is more applicable. The loss ratio method is often used when detailed exposures are not available (Finger, 2001), but the premiums paid are known.

Using the rate changes, the rates can be determined by applying an adjustment factor. The adjustment factor is the experience loss ratio divided by the target loss ratio (McClenahan, 2001). The new rate is calculated as follows:

$$R = AR_0$$

where

R = indicated rate
R_0 = current rate
A = adjustment factor = W/T
W = experience loss ratio
T = target loss ratio

The target loss ratio is calculated by:

$$T = \frac{1 - V - Q}{1 + G}$$

where

V = premium-related expense factor
Q = profit and contingencies factor
G = ratio of non-premium-related expenses to losses

The experience loss ratio is:

$$W = \frac{L}{ER_0}$$

where

L = experience losses
E = experience period earned exposure
R_0 = current rate

Putting these two equations together the final equation is:

$$R = \frac{L(1+G)}{E(1-V-Q)}$$

The loss ratio method is applicable when there is less detailed information or when there are several different sets of relativities. Earned premiums consider the amount paid by different classes, territories, and coverages. However, when earned premiums correspond to historical rate levels, it may be difficult to make adjustments for changes in rate relativities. A disadvantage of the loss ratio method is that it produces only loss ratio and severity relativities. Severity relativities may not be as meaningful if the underlying coverage is not consistent (McClenahan, 2001). As coverages change, the severity of the claims may vary. This must be taken into account before the loss ratio method can be applied. With less detailed information, it may be more difficult to calculate actuarially sound and accurate rates.

Adjustments must be made to the data before applying the loss ratio approach. According to Finger (2001), there are four steps that an actuary must consider:

1. premiums must be adjusted to current rates
2. preliminary adjustments must be made to territorial relativities
3. adjustments are made for credibility

4. the rate change must be balanced for the credibility adjustment

If unadjusted historical earned premiums are used, then relativities are developed for the historical rates rather than the current rates. Some groups of data may be very small, so the actuary must determine the amount of credibility to give the group of data. Credibility is typically based on claim experience, and the current relativity is usually used for the complement of credibility (Finger, 2001).

Losses must be developed to ultimate in both the pure premium method and the loss ratio method since they typically include claims that are not yet settled or claims that are valued on the basis of estimates. Additional adjustments may be made for changes in frequency and severity. Typically the chain ladder, or link ratio, approach is used for developing losses to ultimate. The development of past claims is used to calculate incremental age-to-age development factors. These are then used to calculate age-to-ultimate factors. The losses in each age group are multiplied by the appropriate factors to obtain the estimate of the ultimate loss. This technique may also be called the loss development method.

Ultimate losses can be difficult to calculate since actuaries rely on historical data. The loss development method is based on the assumption that claims go from unreported to reported and unpaid to paid in a pattern consistent with past experience (McClenahan, 2001). Again credibility is important because past experience must be able to be used to predict future development. In addition, some types of claims may take much longer than others to develop. For example, asbestos can take thirty to forty or more years for the claim to be fully developed while an automobile claim for physical damage to a vehicle is known rather quickly. Therefore this claim is paid out and develops to ultimate very quickly. Asbestos has what is known as a long tail and physical damage of cars has as a short tail. In general, property claims develop to

ultimate fairly quickly while liability claims may take longer to be reported and then developed to their ultimate value due to lawsuits and other factors. Claims that have been incurred but have not yet been reported are called IBNR claims. These IBNR claims are estimated and used in actuarial analyses. However, they are very difficult to predict. In addition, claims may develop faster than in the past due to new claim processing systems, increased technology, and many other factors. Companies must consider these and make adjustments to their data. Using development factors is the best way to calculate the ultimate value of the loss.

The actuary must also decide whether to use earned premiums or written premiums. Earned premiums are commonly used since they are the amount of premium collected for the desired amount of time (Finger, 2001). For example, an insured paid $500 for a one year automobile insurance policy. However, only six months have passed, so the earned premium must be calculated for only those six months. In this case, the earned premium used in the analysis is $250. If written premiums were being used then the entire $500 is used in the calculation.

Premiums may also need an adjustment for rate changes since premiums paid by individuals change over time. The adjustments must be made to reflect accurate amounts of premiums (McClenahan, 2001). The historical premiums must be adjusted to the current rate levels to calculate the experience loss ratio. This is done using the parallelogram method. The parallelogram method uses simple geometry to calculate the current rate level and make changes to historical rate levels (McClenahan, 2004). It is also based on the assumption that exposures are written uniformly over the entire calendar period (McClenahan, 2001). Adjusting historical rate levels to the current rate level allows for more accurate analyses. It also creates a consistent base to which factors are compared.

There are other underlying trends that the data must be adjusted for. For example, monetary inflation causes the amount jury's award to increase over time. This increases the cost of the claim to the company and may need to be included in a trend analyses. Generally curves are fitted to the data to help identify trends. Severity and frequency are both commonly affected by various types of trends as well.

The pure premium method and the loss ratio method are most applicable when rates need to be adjusted for a specific group of data. One example might be that automobile insurance for males under 25 exhibits a much higher loss ratio than for all other groups of insureds. All other group's loss ratios are fairly consistent and close to the company's target loss ratio. Therefore, the only loss ratio that needs to be adjusted is the loss ratio for the group, males under 25. In this case, it is best to use the pure premium method or loss ratio method to determine what the real rate should be for this group. It is unnecessary to create an entirely new model for pricing the products since the other groups seem to be in line with the company's goals. Creating a new model takes time and resources, and revising an individual group's rate is a much simpler process. In addition, each time a new rate is created it must be filed with the states where it is changing. The states have to approve these rates before they can be implemented. Changing one specific group's rates may be easier to get the state to approve than changing the whole pricing structure. However, it is possible that the loss ratios could vary widely between groups, and the company could decide it needs an entirely new pricing structure. In this case, data must be collected and analyzed using different procedures to create new rates. The minimum bias procedure and generalized linear models are two processes that can be used to create a structure for pricing products.

11. MINIMUM BIAS PROCEDURE

The minimum bias procedure is used to determine rate relativities simultaneously for two or more classification dimensions. It was first introduced by Robert Bailey and LeRoy Simon in their 1960 Proceedings paper, *Two Studies in Automobile Insurance*. Before the introduction of this new procedure, classification systems were developed by one dimension at a time. This was suitable only in a single dimensional classification system (Feldblum & Brosius, 2003). For example, in a single classification system, the premium might rely only on gender. However, the problem with single classification dimensions is that there are many different classes, each with a different rate. In addition, there is a problem with credibility if there is not a large enough volume of premiums and losses in each class to produce a reliable basis for the rate (Bailey, 1963). Most insurance products today are not accurately priced by the use of single classification dimensions, so the minimum bias procedure has been developed for classification systems with multiple dimensions. For example, in automobile insurance the insured may be rated on gender, age, type of car, past driving history, and various other rating factors. Therefore, classification plans are needed to mathematically combine base rates and rating factors to produce rates for different classes (Mildenhall, 1999). Dr. J. Eric Brosius and Sholom Feldblum combined their own material with the theory of the original actuarial papers by Bailey and Simon to explain the use of the minimum bias.

According to Feldblum and Brosius (2003), the goal of the minimum bias procedure is to create a model that minimizes the difference between the observed loss costs and the indicated pure premiums. Thus, the difference between what the insureds actually pay for their policy and the amount the company has to pay for claims including expenses and profit should be as small

as possible. Once models have been created, bias functions are used to compare them and choose the one that fits the data with the smallest degree of error. Bias is defined as the difference between the observed cost and the fitted value or rate computed by the model.

Bailey and Simon (1960) used four criteria for determining the acceptable set of relativities and making the bias as small as possible. These four criteria were outlined by Bailey and Simon (1960) are:

BaS1: It should reproduce experience for each class and overall (balanced for each class and overall).

BaS2: It should reflect the relative credibility of the various groups.

BaS3: It should provide the minimum amount of departure from the raw data for the maximum number of people.

BaS4: It should produce a rate for each subgroup of risks which is close enough to the experience so that the differences could reasonably be caused by chance.

For condition BaS1, the weighted average rate for each variable must equal the weighted average of the actual experience. Classification rates for each class are balanced and should have a zero bias. Mildenhall (2004) calls this the method of marginal totals. Deviances must be used to determine how well a model fits since according to this condition there is zero bias. The deviance is difference between the observed value and the fitted value according to the model. Deviances for each observation are calculated and totaled to create a measure of model fit. For condition BaS2, the number of exposures in each cell must be taken into account because their variances and credibility may differ. Cells with more observations are considered to have more credibility and are usually weighted heavier. In addition, cells with a large expected variance are

more likely to have large biases and may need to be weighted less than others. These four criteria are important for choosing appropriate rating variables and creating an accurate model.

The minimum bias method has two main models: the multiplicative model and the additive model. The first step is to choose the optimal model. The additive model is used in rare cases. The multiplicative model is used more generally in property and casualty lines of business. Some rating plans use a mixed additive and multiplicative model for creating rates (Fu & Wu, 2007). There are also variations of the multiplicative and additive models that can be used for creating a rating system. Once the model has been chosen, the goodness-of-fit is assessed to determine the appropriateness. The balance principle is most commonly used and the easiest to understand, so it will be explained in this paper.

The goal of the balance principle is to create a model where the sum of the indicated pure premiums equals the sum of the observed loss costs along each row and down each column. Typically, the loss ratio approach is used since many classification systems have uneven distributions. This approach minimizes the potentially distorting effects of other classification dimensions that are not being analyzed in the minimum bias procedure (Feldblum et al., 2003). The process for the following example comes from Feldblum and Brosius' (2003) paper.

The original data must first be split into groups. These groups are usually organized in a matrix form and can be thought of as buckets with groups of data containing similar known characteristics. Below is an example of how the minimum bias procedure is used in practice for automobile insurance. Many different rating factors including age, gender, type of car, driving history, etc. are used in the pricing of automobile insurance. For simplicity, this example will use only two rating factors, age and gender. Insureds are grouped into two age groups: under 25 and over 25. In practice, age would probably be split into more groups, but for demonstration

purposes two groups will be best. After the observed loss costs for each group have been calculated and the premiums are known, the loss ratios can be figured. The loss ratio is the observed loss cost divided by the premium. The following two by two matrix is produced after the loss ratios are calculated:

Table 1: Minimum bias group loss ratios

	Under 25	Over 25	Total
Male	70%	53%	60%
Female	56%	42%	45%
Total	62%	46%	52%

It is evident in Table 1 that the loss ratios for groups of insureds vary. Females over 25 have the lowest loss ratio at 42% and males under 25 have the highest loss ratio at 70%. The ultimate goal is to have consistent loss ratios across all groups of data so that those that are at a higher risk are paying a higher premium and those that are at a lower risk are paying a lower premium. In a perfect classification system, the loss ratios for each group would be equal. In this example, males under 25 are not being charged enough and females over 25 are possibly being charged too much. Companies may strive to have loss ratios equal to their target loss ratio which usually does not include expenses associated with individual policies and profit. In the example, if the company's loss ratio is 52%, then it is shown that males under 25 are not being charged enough and females over 25 are being charged too much. The other two groups have loss ratios fairly close to the target loss ratio, so their rates may not change a significant amount.

There are several causes of uneven loss ratios. First, random loss fluctuations create a credibility issue since there may be claims that are not typical. To fix this, companies can choose to exclude these claims from analyses. One example is catastrophic losses such as hurricanes or tornadoes. These catastrophes typically cause very large losses and can distort the

creation of the pricing plan. Since catastrophes are rare, companies exclude these from analyses and then add a provision for catastrophes after the rate has been calculated. Secondly, uneven loss ratios may be caused by combining improper rate relativities in other classification dimensions with an uneven distribution of insureds. Current rates for a specific group may be too low, and the proportion of insureds paying low rates may be significantly higher or lower in one group than another creating an uneven distribution of insureds between classes. Thirdly, there may be inaccuracies in the rate relativities for the classification dimensions being examined. This cause is usually corrected by the minimum bias procedure. The loss ratio times the current relativity is equal to the relative loss cost. Finally, improper model specification can cause uneven loss ratios. If a multiplicative model is used when an additive model is more appropriate and vice versa, it can lead to uneven loss ratios (Feldblum et al., 2003). For this reason, choosing an appropriate model is very important.

After the loss ratios have been calculated, a base class must be chosen. Normally the class with the lowest loss ratio is chosen as the base class. However, any group can be selected. In this example, females over 25 will be the base class. The base class must be consistent throughout the calculations. It also includes the rate to which factors will be applied.

Next the loss ratio relativities are calculated. The loss ratio relativity is calculated by dividing the loss ratio for each class by the base class loss ratio. After calculation the loss ratio relativities are:

Table 2: Minimum bias loss ratio relativities

	Under 25	Over 25
Male	1.667	1.262
Female	1.333	1.000

The loss ratio relativities in Table 2 show that the loss ratio for males under 25 is 66.7% higher than the loss ratio for females over 25. This is a drastic difference. By adjusting rates, the loss ratios should be more consistent.

It is also important to know the number of exposures in each cell. The exposures are used in adjusting the bias function for the relative amount of business in a particular cell. They assist in making a credibility adjustment based on the absolute volume of business in each cell. More credibility is given to cells with a larger number of exposures and less credibility to those cells with a smaller number of exposures. Depending on the type of insurance, a variety of exposures bases can be used. In automobile insurance, the number of car years in each cell is used as the exposure base. Exposures for each cell are shown in Table 3:

Table 3: Minimum bias number of exposures

	Under 25	Over 25
Male	300	400
Female	600	200

After gathering the required data, the factors for the balance principle are calculated by balancing across the rows and then down the columns. This is continued iteratively until the factors converge to a single number. The following equation is used to calculate the factors:

$$g_i = \frac{\sum_j n_{ij} r_{ij}}{\sum_j n_{ij} a_j}$$

where

g_i: gender group factor
a_j: age group factor
n_{ij}: number of exposures in cell ij
r_{ij}: loss ratio relativity for cell ij

In general, the gender group factor is equal to the sum of the loss ratio relativities for that gender weighted by the number of exposures divided by the sum of the age factors weighted by the number of exposures.

Before solving for the gender factors the first time, there must be a starting point for the age factors. The starting factors can be chosen in a variety of ways. One way is to use one as the starting factor for both of the age groups. More commonly a one way analysis is used to calculate starting factors. The one way analysis calculates the factors for age as if it was the only rating variable. These would be the factors if a single classification dimension was used. Therefore, the over 25 factor is the loss ratio for those over 25 divided by the loss ratio for those under 25 which is equal to 1.348. The factor for those under 25 is one since it is the base class. Using these factors, the gender factors can be found solving the above equation. The gender relativity calculations are:

$$male\ relativity = \frac{(300)(1.667)+(400)(1.262)}{(300)(1.348)+(400)(1.000)} = 1.249$$

$$female\ relativity = \frac{(600)(1.333)+(200)(1.000)}{(300)(1.348)+(400)(1.000)} = 0.991$$

These gender relativities found above are then used to calculate the age relativities and the process continues iteratively. After several iterations the factors in the example converge to the following:

Male (g_1): 1.254
Female (g_2): 1.000
Under 25 (a_1): 1.326
Over 25 (a_2): 1.000

The next step is to calculate the indicated relativities for the model. The indicated relativity is the product of the factors for each group. For example, the male under 25 indicated relativity is equal to the male relativity (1.254) times the under 25 relativity (1.326). The table below shows the indicated relativities for the example.

Table 4: Minimum bias indicated relativities

	Under 25	Over 25
Male	1.662	1.254
Female	1.326	1.000

The example in this paper shows that the rate for males under 25 should be 66.2% more than the rate for females over 25 to make loss ratios more consistent across each group. According to past experience, charging males under 25 66.2% more, females under 25 32.6% more, and males over 25 25.4% more than the base class of females over 25 should give a more even distribution of the loss ratios for the entire population. The base rate is chosen and the factors are applied to the base rate to get the rates for each group.

Determining how well the model represents the data is important after the model has been developed. In statistics, the best estimate is the unbiased estimator with the least variance. For mathematical distributions (i.e. normal distribution, Poisson distribution), the mean has several unbiased estimators. The problem is determining which unbiased estimator has the least variance, and therefore, is most reliable. In insurance statistics, however, there are not many unbiased estimators to choose from. According to Bailey (1963), combining all classes to produce a single adjustment gives an unbiased sample mean, but none of the resulting rates are unbiased for each class. Thus, subdividing the data allows resulting rates to have less bias in each class. It has not been proven that combining adjustments gives an unbiased rate, so the biggest problem is to find the estimator with the least bias. After several sets of rates are

calculated, it must be determined which set is the least bias. Using the average absolute difference between the estimated rates and the data for each multiple subdivision of the data is the best way to do this. The differences are weighted by the number of risks or the amount of premium in each subdivision to account for credibility. The sums of squares differences also may be sufficient for determining which set of rates has the least bias (Bailey, 1963).

In addition to the balance principle, several other minimum bias methods have been developed. One is to minimize the average absolute error between the indicated and observed figures. This method is rarely used in insurance. The average absolute error can be easily calculated, but it may be more difficult to find the set of relativities that minimizes it. A second method is to minimize the sum of squared errors between the indicated and observed loss costs. The squared error for each cell is equal to the square of the difference between the observed loss costs and the indicated pure premium. The partial derivatives of these equations are taken and set equal to zero to solve for the classification relativities. This method once again uses an iterative process to find the final relativities. In the sum of squared error method, the total squared error is minimized while the balance principle ensures that the error in each classification dimension is zero. The formula for this model is:

$$x_i = \frac{\sum n_{ij} r_{ij} y_j}{\sum n_{ij} y_j^2}$$

There are also two other models that can be used for the minimum bias procedure. These include the chi-squared error and maximum likelihood estimation. All four of these models are discussed with further detail in *The Minimum Bias Procedure: A Practitioner's Guide* by Sholom Feldblum and Dr. J. Eric Brosius.

The minimum bias method relies strictly on arithmetic functions. However, the data in the minimum bias method may be organized in rows and columns, but it does not imply that there is an arithmetic relationship. Venter (1990) has developed two methods in response to this criticism. The first is to allow individual cells to vary from the arithmetic function. The second is to estimate individual cells without using an arithmetic function by using credibility methods.

In practice, the minimum bias procedure can be applied to a variety of models to further refine classification systems. However, one problem is that the method does not allow an actuary to determine which variables are statistically significant. They lack statistical framework which prevents actuaries from testing for statistical significance which can lead to insignificant variables being placed in the model. Therefore, variables with low statistical significance may be included in the model while those with high statistical significance may be left out.

The minimum bias procedure poses another disadvantage because it requires a lot of detailed information for the procedure to work. For example, in automobile insurance age, gender, location, and varying other statistics may be necessary to develop a classification system for pricing policies. One way to determine if a variable is a good predictor of losses is to examine how the loss ratio changes in the group. First, the data is divided into at least four equal groups for a given variable. For example, if age is the variable, the data might be divided into age 16-25, 26-40, 41-60, and 60+. It is important to have a fairly even distribution of premiums in each of the groups. Then the loss ratios are calculated for each group. The loss ratios are examined to see if there is a significant increase or decrease from one group to another. If the loss ratios appear to be similar, then the minimum bias method is not useful. However, if there is a significant increase or decrease in the loss ratios, then the minimum bias procedure may be useful. This can be done for a variety of variables to help in deciding which ones to use in the

classification system. An advantage of the minimum bias procedure is that multiple variables can be used to price insurance products more efficiently and effectively based on group characteristics. Once the variables have been determined, they can be incorporated into the pricing plan. There is no proof that these variables are statistically significant, but according to the minimum bias analysis and the loss ratio observations, they seem to be good predictors for losses.

Another problem may be that rating plans should be a mixed additive and multiplicative model. However, most software programs do not allow for use to the two models together. Lastly, parameters calculated by the minimum bias model may be questionable or unacceptable in the marketplace (Wu et al., 2007). The actuary may have to constrain the factors based on business and competitive reasons which could reduce the accuracy of the model. This problem has to do with the legal constraints that actuaries and insurance companies face when creating pricing plans. It could be illegal to price based on a certain characteristic in some locations. There also may be restrictions on the amount of increase or decrease in price due to a certain characteristic in some locations.

Predictive modeling is becoming more popular in property and casualty insurance. The minimum bias procedure is one model that can be used for predictive modeling. While the example in this paper only shows the use for a two by two case, the procedure can be extended for multiple dimensions and many different groups. However, the groups must have enough data to be credible. If the groups are too small, it will be hard to get an accurate prediction of what the premium should be. With the use of technology the factors can be calculated fairly easily using an iterative process. Insurance products such as automobile insurance and homeowners insurance are using much more predictive modeling than in the past. These products both have a

large variety of data that can be collected, and thus, incorporated into the model. The biggest challenge lies in determining which rating variables are a good predictor of losses and can be used to create the best rating plan, or model.

The minimum bias model is very easy to use and apply in practice. The calculations of relativities are straightforward. The group that the individual falls into determines its rate. The relativities are applied to the base rate to get the premium for the individual. While the minimum bias model lacks a statistical framework, it is still used in practice because of the ease of calculations. The minimum bias method may be used for a starting point in creating a classification plan, or it could be used to calculate specific prices.

12. GENERALIZED LINEAR MODELS

Many of the common minimum bias models are a special case of generalized linear models (GLMs). Generalized linear models have a statistical framework and allow assumptions to be made about the relationships with predictive variables as well as the nature of the data itself. Also, they are solved more efficiently since they are not solved using an iterative approach. The variables that are statistically significant can be determined because of the statistical evaluation of the variables. Model assumptions can also be validated (Anderson et al., 2005). In insurance today, generalized linear models are used to price private passenger auto and other personal lines as well as small commercial lines particularly in the European Union. The goal is to use a model to replace the data with a smaller set of parameters for calculating the proposed rates. Generalized linear models can be used for ratemaking and underwriting. The

generalized linear model process described below is described by Anderson et al. (2005) in *A Practitioner's Guide to Generalized Linear Models.*

The goal of generalized linear models is to determine the relationship between an observed response variable, Y, and a variety of predictor variables (covariates), X. The linear models are written as Y equals the sum of its mean, μ, and a random variable, ε.

$$Y = \mu + \varepsilon$$

The assumption is that μ, the expected value of Y, can be written as a linear combination of the predictor variables, X. The error term, ε, is normally distributed with mean zero and variance σ^2.

Using data from the paper *Minimum Bias with Generalized Linear Models* by Brown (1988) we get the following table:

Table 5: Average loss costs

	Under 25	Over 25
Male	399.53	183.40
Female	243.45	153.02

The data in the paper is grouped into classes based on characteristics of automobile insurance. Age, gender, and urban or rural are coded into the data set. Class 7 from the paper was used for the calculation of losses for males over 25. Classes 8 through 13 were used to calculate the average loss costs for males under 25. The average loss cost for females over 25 used class 2. Lastly, average loss costs for females under 25 used classes 18 and 19. Urban and rural classifications were added together to find the loss cost for each group.

This results in having four predictor variables, male (X_1), female (X_2), under 25 (X_3), and over 25 (X_4). These are indicator variables that take the value one or zero. If the age of each individual was known, they could be used in the calculations rather than grouping the data into

two groups. However, for this example, there are only two groups for the age. The goal of the linear model is to find parameters (β_1, β_2, β_3, and β_4) for the following equation:

$$Y = \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3 + \beta_4 X_4 + \varepsilon$$

Using this model, there is the same number of parameters as rating factor levels to be considered which creates a linear dependency between the four covariates. Therefore, the model does not have unique parameters. The following model can be considered to get unique parameters:

$$Y = \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3 + \varepsilon$$

This model assumes that there is an additive effect for those under 25, which is the same regardless of gender. In addition, this model may be thought of as having a base class of female over 25 with addition effects for being a male (β_2-β_1) and under 25 (β_3).

The observations from the example can be expressed as a system of equations:

$$\begin{aligned}
399.53 &= \beta_1 + 0 + \beta_3 + \varepsilon_1 \\
183.40 &= \beta_1 + 0 + 0 + \varepsilon_2 \\
243.45 &= 0 + \beta_2 + \beta_3 + \varepsilon_3 \\
153.02 &= 0 + \beta_2 + 0 + \varepsilon_4
\end{aligned}$$

In the classical linear model, the parameters are selected by minimizing the sum of squares errors (SSE):

$$\begin{aligned}
SSE &= \varepsilon_1 + \varepsilon_2 + \varepsilon_3 + \varepsilon_4 \\
&= (399.53 - \beta_1 - \beta_3)^2 + (183.40 - \beta_1)^2 + (243.45 - \beta_2 - \beta_3)^2 + (153.02 - \beta_2)^2
\end{aligned}$$

The derivatives are then taken with respect to each of the parameters and set equal to zero. The set of equations are then solved to get the final factors. The final factors are:

$\beta_1 = 214.83$
$\beta_2 = 121.60$
$\beta_3 = 153.28$

These numbers are substituted into the equation to form the rating plan. The base rate for males is $214.83 and the base rate for females is $121.60. There is an additional $153.28 charged for those individuals under the age of 25. The equation becomes:

$$Y = 214.83X_1 + 121.60X_2 + 153.28X_3 + \varepsilon$$

These equations can also be solved using vector notation. This creates a more compact form and is an easier way of evaluating the parameters for the model. This is especially useful as the number of observations and the number of predictor variables increases. $\underline{Y}$ is the column vector which corresponds to the response variable observed values:

$$Y = \begin{vmatrix} 399.53 \\ 183.40 \\ 243.45 \\ 153.02 \end{vmatrix}$$

Generalized linear models use probability distributions for an observed random variable given a set of rating factors. According to Robert Brown (1988) in his paper *Minimum Bias with Generalized Linear Models*, these explanatory vectors must satisfy three conditions:

1. The distribution of the random variable given the explanatory vectors belongs to the exponential family.

2. The explanatory vectors enter only as a linear sum of their effects.
3. The expected value of each observation can be calculated as a function of the linear sum of effects.

X is the matrix containing the predictor variables for the observed values:

$$X = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

$\underline{\beta}$ is the column vectors of parameters and $\underline{\varepsilon}$ is the column vector of residuals:

$$\underline{\beta} = \begin{bmatrix} \beta_1 \\ \beta_2 \\ \beta_3 \end{bmatrix} \qquad \underline{\varepsilon} = \begin{bmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \varepsilon_3 \\ \varepsilon_4 \end{bmatrix}$$

The equation now takes the form:

$$\underline{Y} = X\underline{\beta} + \underline{\varepsilon}$$

The values of the components of $\underline{\beta}$ is calculated to get the parameters for the model. $\underline{\beta}$ minimizes the sum of squares of the components of $\underline{\varepsilon}$. Therefore, the components of the linear model are the set of assumptions about the link between $\underline{Y}$ and the predictor variables and the objective function that is optimized in order to solve the equations.

It is assumed in linear models that all observations are independent and come from a Normal distribution. This can create a problem if the Normal distribution assumption is not valid. It is also assumed that the mean is a linear combination of the predictor variables and each

component of the random variable has a common variance. Therefore, McCullagh and Nelder (1989) outline assumptions for the linear model:

1. Random component: The components of $\underline{Y}$ are independent and normally distributed. The mean of each component has a common variance.
2. Systematic component: The "linear predictor," η, is a combination of the predictor variables.

$$\underline{\eta} = X\underline{\beta}$$

3. Link function: The link function specifies the relationship between the random component and the systematic component. For the linear model the link function is the same as the identity function.

$$E(Y) = \underline{\mu} = \underline{\eta}$$

Linear models have several limitations. First, assuming normality and constant variance for response variables may not always be valid. Therefore, the data may need to be transformed to use these assumptions. Second, response variable values might only be positive violating the assumption of normality. If the response variable is non-negative, the variance is a function of the mean because the variance of Y tends to zero as the mean of Y tends to zero. Finally, the additivity of effects along with the second and third assumptions may not be realistic for a variety of applications (Anderson et al., 2005). For example, it may be more accurate to use the multiplicative property than the additive property to calculate the response variable. Many insurance risks tend to use the multiplicative model. However, using a multiplicative property violates the assumptions of the linear model.

For generalized linear models the linear model assumptions of normality, constant variance, and additivity of effects are no longer used. The response variable is assumed to be a

member of the exponential family. The variance can vary with the mean of the distribution. The effect of the predictor variables is assumed to be additive on a transformed scale. Therefore, the generalized linear model assumptions according to Anderson et al. (2005) are:

1. Random component: The components of $\underline{Y}$ are independent and come from an exponential family of distributions.
2. Systematic component: The "linear predictor," η, is a combination of the predictor variables.

   $$\underline{\eta} = X\underline{\beta}$$

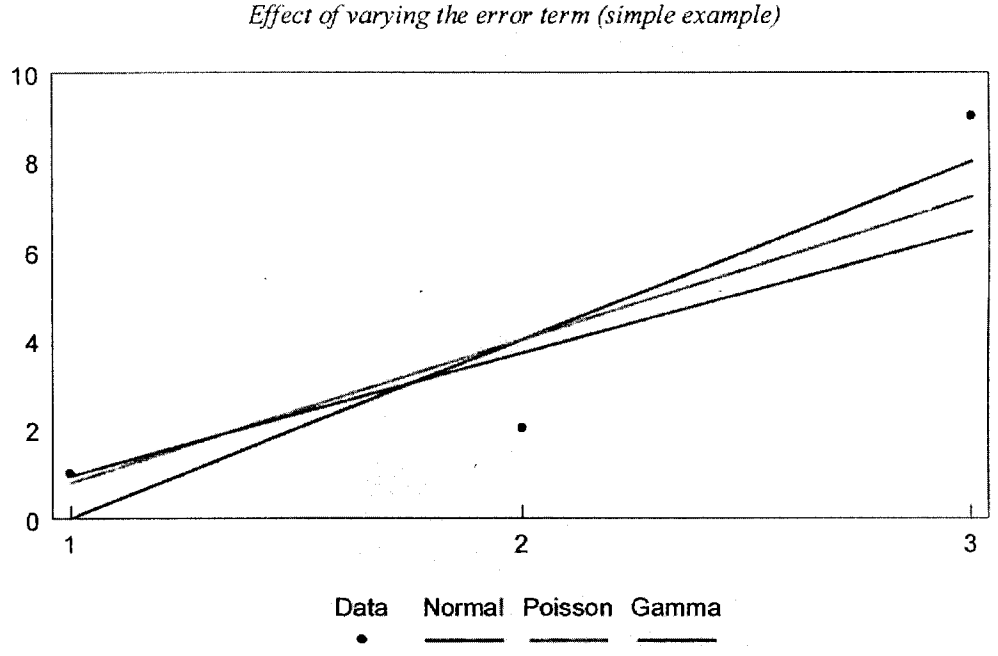
3. Link function: The link function specifies the relationship between the random component and the systematic component. The link function, g, is differentiable and monotonic.

   $$E(\underline{Y}) = \underline{\mu} = g^{-1}(\underline{\eta})$$

Members of the family of exponential distributions have the following properties:

1. the distribution is completely specified in terms of its mean and variance
2. the variance of Y_i is a function of its mean

Figure 1: Error term analysis for generalized linear models



Graph taken from page 15 of *A Practitioner's Guide for Generalized Linear Models* (Anderson et al., 2005).

The effects of the error terms were graphed by Anderson et al. (2005) to show that the distribution of the error term changes the fitted values of the model. When the Normal, Poisson and Gamma variance functions are considered it can be seen that the fitted values from the normal variance function are attracted to the original data points with equal weight. For the Poisson model, the fitted values with smaller expected value influence the model more than the fitted values with high expected value, and thus, a higher assumed variance. In the gamma variance function, it is assumed that the variance increases with the square of the expected value. Therefore, the points with the smaller expected value influence the model even more than those with the larger expected value.

Prior weights of each observation are incorporated into the model for credibility purposes. The exposure of each observation is defined by w_i. The observations with higher exposure have a lower variance than those with lower exposure. This means that the model will

be more influenced by those observations that have a higher exposure. Using prior weights, lower credibility can be attached to parts of the data that are known to be less reliable.

Generalized linear models require approximately 100,000 exposures for the model to be fully credible. Smaller volumes of exposures can be used in some cases, but the data may need to be weighted according to volume. Data should also be based on at least two years of claim experience. This prevents the model from being distorted by events during the year. Combining geographical areas can also help in creating better models. However, when geographic areas have significantly different models, combining geographic regions may produce incorrect results.

Data sets for generalized linear models need to be detailed just like data sets for the minimum bias models. Generalized linear model data sets must include premium data, earned exposures, incurred loss amounts, and the number of incurred claims. These should be split into claim types to allow for more in depth analyses. It is also necessary to have the raw explanatory variables and dummy variables. Explanatory variables include individual risk characteristics such as gender, age, etc. If the detailed data set is difficult to obtain, the generalized linear model approach may be impractical.

Generalized linear models are more efficient than the traditional minimum bias approach because iterative processes do not have to be programmed to obtain the parameters (Wu et al., 2007). Powerful computers are making this issue less significant since they are able to do a lot of calculations in a very small amount of time. However, the software programs for generalized linear models are able to create pricing plans in a much quicker and efficient way since the models are already programmed into them.

Generalized linear models may fall prone to model-selection limitation. They assume that the underlying distribution is from the exponential family (Wu et al., 2007). Some statistical

software packages may even limit this further. If the exponential family assumption is not valid, then the rating plan will be invalid and inaccurate. The model-selection limitation may also reduce accuracy of the pricing plan since data is not usually perfect. Models from the exponential family may not fit the data well. However, assumptions are made in any pricing plan. If a reasonable assumption can be made, then actuaries must use it to determine the pricing model.

The generalized linear model approach allows actuaries to determine which variables are statistically significant. This is an advantage over the use of the minimum bias procedure. Using computer software programs such as SAS to create models allows actuaries to determine which variables are significant. Therefore, only statistically significant variables are used in the model. In addition, the use of statistical models allows the model fit to be determined. This is very difficult in the minimum bias procedure. There are a variety of generalized linear models that can be compared to determine which model has the best fit.

13. CONCLUSION

Each of the methods discussed in this paper is useful in different situations. The pure premium is used to update premiums based on changes in variable or fixed costs. It is most valuable when earned exposures are used in the calculation. The loss ratio method is similar to the pure premium method but uses earned premium in its calculations rather than earned exposures. In addition, it calculates a rate relativity that can be applied to the current rate to determine the new rate. These two methods are not used to develop a classification system with

many rating variables. They are unable to consider data grouped by characteristics in determining rates for specific groups.

The minimum bias procedure and generalized linear models are used to create classification plans based on the characteristics of groups and individuals. Both of these models can incorporate a variety of rating variables to construct more refined classification plans. These procedures are flexible and provide for addition and deletion of rating variables over time. The flexibility of these models is useful in the dynamic insurance environment. There are new rating variables introduced to insurance companies occasionally that can be incorporated into models to provide for better and more accurate pricing. The minimum bias procedure and generalized linear models also provide a rather easy way for determining rates for an individual. The individual's characteristics are run through a model to determine the premium they should be charged. These models are very useful for agents and brokers since the model can specify rates for nearly any combination of characteristics.

The minimum bias procedure uses a matrix to create relationships between rows and columns. The main problem with this is that not all rating variables have an arithmetic relationship. For example, there may not be an arithmetic relationship to determine how much more a rural person should pay for automobile insurance than an urban person. In addition, the statistical significance of the rating variables is unknown. Insignificant rating variables could be included in the model while significant rating variables could be left out. The minimum bias procedure also requires a large data set for analysis. Data sets must be very detailed for the rate implementation to be effective and efficient. It may be more practical to determine which rating variables can be used to apply credits and debits to insurance premiums rather than exactly how much the premium should be credited or debited. With developing technology, the minimum

bias procedure is becoming easier to calculate because of increasing computer power. However, there may be better models for determining the rates for individuals based on group characteristics.

Generalized linear models use a statistical approach for calculating insurance rates. Using generalized linear models allows an actuary to determine which rating variables are of statistical importance. Only those rating variables that are statistically significant are included in the model. There are a variety of models that can be created and compared to find the best model for the insurance product. Technology has allowed generalized linear models to become more useful in insurance companies since the model fit can be determined. In comparison to the minimum bias model, generalized linear models allow companies to determine how well the model fits and thus the accuracy and appropriateness of the pricing plan.

Both the minimum bias model and generalized linear models can be used when the data set is very detailed and rates are needed for multi dimensional classification systems. However, determining which one is better may be more difficult. The minimum bias model is much simpler and easier to understand. It can be applied to data sets very easily. It may be more useful to determine which rating variables appear to have an effect on the risk of the insured rather than exactly how much effect. The minimum bias procedure can be used to determine which specific groups should be credited or debited and then incorporated into a checklist to determine how much they are credited or debited. Generalized linear models may be more useful in determining exactly how much an insured should be charged based on their individual characteristics. While they are more difficult to implement, they can also be tested using statistical procedures. This allows actuaries to determine the accuracy of their model. As technology continues to develop, generalized linear models are going to become more useful.

Statistical packages will be able to calculate and compare the models quickly so that actuaries can determine exactly how much an insured should be charged.

This paper has described a variety of pricing models which can be used in pricing insurance products. Each of them has their own advantages and disadvantages. The loss model approach and the pure premium approach require very little data while the minimum bias procedure and generalized linear model require detailed data sets. However, the minimum bias procedure and generalized linear models are more useful for determining rates based on individual risk characteristics. With the increase of predictive modeling for insurance products today, the minimum bias procedure and generalized linear models are gaining acceptance. These methods allow actuaries to create more refined classification systems that can be used to more accurately price insurance products. The minimum bias procedure and generalized linear models will be further developed and assist actuaries in creating accurate pricing plans.

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